MODULAR KOSZUL DUALITY: SECTIONS 5–8

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Abstract. Sections 5–8 develop the collision residue and its consequences. Section 5 extracts the seven faces of the classical r-matrix: bar–cobar twisting morphism, DNP 25 line-operator generator, Khan–Zeng PVA descent, Gaiotto–Zeng sphere Hamiltonians, the Drinfeld Yangian, the Sklyanin/STS 83 realisation, and the FFR 94 Gaudin limit. The seven faces are projections of the single R-matrix carried by the ordered bar; the symmetric bar retains only the scalar shadow κ. Section 6 records the collision residue in the pole-order dichotomy, separating families with binary, ternary, and higher collisions. Section 7 installs the shadow obstruction tower with quadrichotomy $G/L/C/M$ and universal coefficient growth $C_{\mathcal{A}} = 6$ on non-logarithmic class M with Virasoro subalgebra. Section 8 traces the arithmetic of the shadow L-object, records the characteristic Kummer-irregular primes $\{691, 3617\}$ present through $r = 11$, and specifies that the primes $\{1423, 3067, 23, 43, 419\}$ appearing in S_r numerators are Riccati-arithmetic characteristic primes only, not Kummer-irregular at the primary source.

1. The collision residue and the seven faces

The bar differential extracts OPE data at collision divisors. When two points collide, the residue of the bar-intrinsic MC element $\Theta_{\mathcal{A}}$ at the codimension-one stratum $D_{\{1,2\}} \subset \overline{\mathrm{FM}}_2(X)$ defines the *collision residue*:

$$r_{\mathcal{A}}(z) \ := \ \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \ \in \ \mathrm{End}_{\mathcal{A}}(2)[[z^{-1}]]\,.$$

The collision residue is the degree-2, genus-0 projection of $\Theta_{\mathcal{A}}$. Its pole orders are one less than those of the OPE: the logarithmic kernel $d\log(z-w)$ absorbs one power of $(z-w)$. For a chiral algebra with OPE poles of maximal order p, the collision residue has poles of order $p-1$.

1.1. Pole-order dichotomy.

The maximal pole order of the collision residue separates the standard landscape into two regimes:

Pole order	Algebra	Collision	Class
1	Heisenberg $\mathcal{H}_k$	k/z	**G**
1	Kac–Moody $\widehat{\mathfrak{g}}_k$	$\Omega/((k+h^\vee)z)$	**L**
3	Virasoro Vir_c	$(c/2)/z^3 + 2T/z$	**M**
$2N-1$	$\mathcal{W}_N$	poles through $z^{-(2N-1)}$	**M**

This table encodes a structural dichotomy: algebras whose collision residue has at most a simple pole (classes **G** and **L**) have finite shadow depth; only class **G** is Swiss-cheese-formal on the derived-center side. Algebras with higher-order poles (class **M**) have infinite shadow towers and genuinely non-formal A_∞-structure.

The dichotomy is not a property of individual OPE coefficients but of the collision residue as a whole: the $d\log$ absorption converts the OPE's double pole (carrying the metric) and simple pole (carrying the Lie bracket) into a single simple pole (carrying the Casimir tensor) for Kac–Moody, while

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the Virasoro OPE's quartic pole becomes a cubic collision pole, producing an r-matrix with genuine spectral dependence.

1.2. **Seven faces of $r(z)$.** The collision residue is a single object viewed through seven algebraic lenses.

Theorem 1.1 (Seven-face identification). *For a modular Koszul chiral algebra $\mathcal{A}$, the following seven expressions are equal as elements of* $\mathrm{End}_{\mathcal{A}}(2)[[z^{-1}]]$*:*

- (F1) *The bar collision residue:* $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$.
- (F2) *The spectral R-matrix of the ordered bar:* $R(z) = \mathrm{id} + r(z)/z + \cdots$ *solves the classical Yang–Baxter equation.*
- (F3) *The λ-bracket of the Poisson vertex algebra:* $r(z) = \int_0^\infty e^{-\lambda z}\{a_\lambda b\}\, d\lambda$ (*Laplace transform*).
- (F4) *The Knizhnik–Zamolodchikov connection:* $\nabla_{\mathrm{KZ}} = d - r(z)\, d\log(z_1 - z_2)$.
- (F5) *The Drinfeld r-matrix* (*for* $\mathcal{A} = \widehat{\mathfrak{g}}_k$)*:* $r(z) = \Omega/((k+h^\vee)z)$*. Equivalently, in the trace-form convention* $r(z) = k\,\Omega_{\mathrm{tr}}/z$ *with* $k\,\Omega_{\mathrm{tr}} = \Omega/(k+h^\vee)$ *at generic k.*
- (F6) *The Sklyanin bracket:* $\{L_1(u), L_2(v)\} = [r(u-v), L_1(u) \otimes L_2(v)]$.
- (F7) *The Gaudin Hamiltonians:* $H_i = \sum_{j\neq i} r(z_i - z_j)$ *are mutually commuting.*

Faces (F1)–(F4) *hold for every modular Koszul algebra. Faces* (F5)–(F7) *specialize to affine Kac–Moody; the general case is obtained by replacing* $\Omega/((k+h^\vee)z)$ *with* $r_{\mathcal{A}}(z)$*.*

The seven faces are not seven theorems but seven readings of one equation. The collision residue $r(z) = \pi_2(\Theta_{\mathcal{A}})$ is the degree-2 projection of the universal MC element; the seven faces are the seven classical frameworks in which this projection has been studied. Drinfeld's r-matrix [?Drinfeld85], the KZ connection [?KZ84], the Sklyanin bracket [?STS83], and the Gaudin model [?FFR94] all extract the same binary invariant.

The relationship to the modular characteristic is immediate:

$$\kappa(\mathcal{A}) \;=\; \mathrm{av}\big(r(z)\big) \quad \text{for abelian and Virasoro-type families,}$$

The averaging map av is the Σ_2-coinvariant projection. For non-abelian affine Kac–Moody in the trace-form convention,

$$\mathrm{av}(k\,\Omega_{\mathrm{tr}}/z) \;=\; \frac{k\ \dim(\mathfrak{g})}{2h^\vee} \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k), \qquad \kappa(\widehat{\mathfrak{g}}_k) \;=\; \kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) + \frac{\dim(\mathfrak{g})}{2}.$$

For Virasoro, $\mathrm{av}((c/2)/z^3 + 2T/z) = c/2$.

1.3. **DS reduction as a functor on collision residues.** Drinfeld–Sokolov reduction converts affine Kac–Moody algebras into $\mathcal{W}$-algebras: $\widehat{\mathfrak{g}}_k \xmapsto{\mathrm{DS}} \mathcal{W}^k_N(\mathfrak{g})$. At the level of the collision residue, this passage raises the pole order:

	OPE max. pole	**Collision max. pole**	**Class**
$\widehat{\mathfrak{sl}}_N$	2	1	**L**
$\mathcal{W}_N$	$2N$	$2N-1$	**M**

The Sugawara construction generates a Virasoro field $T(z)$ from the affine currents $J^a(z)$. The OPE $T(z)\,T(w)$ has a quartic pole; DS reduction extends this mechanism to all generators, producing a $\mathcal{W}$-algebra whose maximal OPE pole grows with N. The shadow depth jumps from finite (class **L**) to infinite (class **M**): DS reduction is the functor that transports gauge-type algebras into gravitational-type algebras.

The intertwining theorem controls the Koszul-dual behavior: the bar-cobar adjunction commutes with principal DS reduction. For non-principal reductions (hook-type in type A, proved; arbitrary nilpotent, programme), the intertwining holds on the hook-type corridor and is conjectural in full generality.

2. The Swiss-cheese realization

The bar complex of a chiral algebra on a curve X has two structures: a differential (from OPE residues on Fulton–MacPherson compactifications of X) and a coproduct (from interval-cutting in a topological direction). The bar complex $B(\mathcal{A})$ is an E_1 chiral coassociative coalgebra over $(\mathrm{ChirAss})^!$: the differential encodes holomorphic factorization on $\mathbb{C}$, the deconcatenation coproduct encodes topological factorization on $\mathbb{R}$. The two-coloured Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$ emerges on the chiral derived center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A})$, not on $B(\mathcal{A})$ itself.

2.1. The two colours. The bar complex lives on $\overline{\mathrm{FM}}_k(\mathbf{C}) \times \mathrm{Conf}_k(\mathbb{R})$, the product of holomorphic and topological configuration spaces. This product is the operadic fingerprint of the Swiss-cheese operad:

	Closed colour	**Open colour**
Geometry	$\overline{\mathrm{FM}}_k(\mathbf{C})$	$\mathrm{Conf}_k(\mathbb{R})$
SC datum	$C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$	$\mathcal{A}$
Bar engine	Collision-residue differential on $B(\mathcal{A})$	Deconcatenation coproduct on $B(\mathcal{A})$
Physics	Bulk operators	Boundary operators

The bar differential and the deconcatenation coproduct are the two operations of the ordered E_1 dg coalgebra $B(\mathcal{A})$. They record the holomorphic and topological directions, but they do not by themselves furnish the two colours of a Swiss-cheese algebra. The closed and open colours live on the derived-center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A})$: bulk operators act on the boundary, but not conversely.

The directionality of the Swiss-cheese operad is strict: *no open inputs produce closed outputs*. Bulk operators restrict to boundary operators, but not conversely. The open colour is the E_1-direction; the passage to Σ_n-coinvariants (the closed/symmetric bar) is a lossy projection. The five main theorems A–H are the invariants that survive this projection.

2.2. Three bar complexes. Three bar-type objects coexist, corresponding to three cooperadic structures on the same underlying graded vector space:

(a) $\overline{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$: the *ordered bar*. Deconcatenation coproduct ($n+1$ terms), coassociative but not cocommutative. Carries the R-matrix, the KZ associator, and the full Yangian deformation. This is the E_1-coalgebra.

(b) $\overline{B}^{\Sigma}(\mathcal{A}) = \mathrm{Sym}^c(s^{-1}\bar{\mathcal{A}})$: the *symmetric bar*. Coshuffle coproduct (2^n terms), cocommutative and coassociative. This is the factorization coalgebra on $\mathrm{Ran}(X)$: the E_∞-coalgebra of Beilinson–Drinfeld.

(c) $B^{\mathrm{FG}}(\mathcal{A}) = \mathrm{Lie}^c(s^{-1}\bar{\mathcal{A}})$: the *Francis–Gaitsgory bar*. CoLie cobracket, capturing only the zeroth-pole OPE data ($a_{(0)}b$, the chiral Lie bracket). A quasi-isomorphism $B^{\mathrm{FG}} \hookrightarrow \overline{B}^{\Sigma}$ holds in characteristic zero, but B^{FG} is invisible to the curvature $\kappa(\mathcal{A})$: the modular characteristic comes from the *first* OPE pole, which the Francis–Gaitsgory bar does not see.

The natural descent is $\overline{B}^{\mathrm{ord}} \xrightarrow{\mathrm{av}} \overline{B}^{\Sigma} \hookleftarrow B^{\mathrm{FG}}$: the ordered bar maps to the symmetric bar by Σ_n-coinvariance (a lossy projection), and the Francis–Gaitsgory bar embeds into the symmetric bar (a quasi-isomorphism in characteristic zero, losing all higher-pole data). The ordered bar is the natural primitive; the other two are its images.

2.3. **PVA descent and the recognition theorem.** The cohomology of a Swiss-cheese algebra carries a Poisson vertex algebra structure: the commutative product descends from the regular OPE and the Poisson bracket from the singular OPE. This is the *PVA descent*: the classical shadow of the full quantum Swiss-cheese structure.

The six PVA axioms (Leibniz, skew-symmetry, Jacobi, sesquilinearity, ∂-bracket, and Wick) are proved geometrically from configuration-space geometry: the exchange cylinder on $\overline{\mathrm{FM}}_2(\mathbf{C})$ for skew-symmetry, the three-face Stokes theorem on $\overline{\mathrm{FM}}_3(\mathbf{C})$ for Jacobi, and boundary factorization on higher $\overline{\mathrm{FM}}_k(\mathbf{C})$ for the remaining axioms.

The recognition theorem establishes the converse:

Theorem 2.1 (Recognition)**.** *A factorization algebra on* $\mathrm{Ran}(X)$ *equipped with Weiss cosheaf descent data and compatible with the factorization isomorphisms is a* $C_*(\mathcal{W}(\mathrm{SC}^{\mathrm{ch,top}}))$*-algebra.*

The theorem converts the abstract factorization structure into a concrete operadic algebra over the Boardman–Vogt resolution of the Swiss-cheese operad. It is proved by Weiss cosheaf descent: the factorization pattern on $\mathrm{Ran}(X)$ determines the operadic composition uniquely.

The homotopy-Koszulity of $\mathrm{SC}^{\mathrm{ch,top}}$ is proved via Kontsevich formality and transfer from the classical Swiss-cheese operad (Livernet). This removes all previously conditional hypotheses: bar-cobar Quillen equivalence, filtered Koszul duality, and the identification of line operators with dual modules all become unconditional.

3. Three-dimensional HT QFT and gauge-gravity

A four-dimensional holomorphic-topological field theory on $\mathbf{C}_z \times \mathbb{R}_t \times \mathbb{R}_{>0}$ restricts to a chiral algebra on the holomorphic boundary. The bar complex of that boundary algebra classifies twisting morphisms (couplings to the Koszul dual); the chiral derived centre computes the universal closed sector; bar-cobar inversion recovers the boundary algebra itself. The modular convolution algebra organizes the full boundary-to-bulk correspondence at all genera.

3.1. **The holographic datum.** The complete data of a holographic system is a sextuple:

Definition 3.1 (Holographic modular Koszul datum)**.** The *holographic modular Koszul datum* of a chiral algebra $\mathcal{A}$ is the sextuple

$$\mathcal{H}(\mathcal{A}) \;=\; \big(\mathcal{A},\ \mathcal{A}^!,\ \mathcal{C},\ r(z),\ \Theta_{\mathcal{A}},\ \nabla^{\mathrm{hol}}\big),$$

where $\mathcal{A}^! = (H^*(B(\mathcal{A})))^\vee$ is the Koszul dual, $\mathcal{C} \simeq \mathcal{A}^!\text{-}\mathsf{mod}$ is the line-operator category, $r(z)$ is the collision residue, $\Theta_{\mathcal{A}}$ is the universal MC element, and $\nabla^{\mathrm{hol}}_{g,n} = d - \mathrm{Sh}_{g,n}(\Theta_{\mathcal{A}})$ is the modular shadow connection. Every component is a projection of $\Theta_{\mathcal{A}}$.

The datum $\mathcal{H}(\mathcal{A})$ is algorithmically reconstructed from the boundary OPE of $\mathcal{A}$: $\mathcal{A} \overset{B}{\longmapsto} \Theta_{\mathcal{A}} \xmapsto{\mathrm{Res}^{\mathrm{coll}}} r(z) \xmapsto{\mathrm{RTT}} \mathcal{C}$. Each step uses only the output of the preceding step; the entire holographic system is determined by the chiral algebra on the boundary.

3.2. **The 4d–3d–2d resolution.** The three volumes of the programme correspond to three levels of a dimensional hierarchy:

d	**Level**	**Operadic type**	**Key datum**	**Volume**
4	CY source	E_2	CY category C	III
3	Swiss-cheese	E_1	R-matrix $r(z)$	II
2	Modular shadow	E_∞	$\kappa(\mathcal{A})$	I

At each level, information is lost. From 4d to 3d, one topological direction is forgotten: $E_2 \to E_1$ (the braiding becomes ordering). From 3d to 2d, Σ_n-coinvariance projects the R-matrix to a scalar: $r(z) \mapsto \kappa$. The five main theorems are the invariants that survive both projections.

3.3. **Gauge versus gravity.** The shadow depth classification (Definition **??**) separates the standard landscape into two physical regimes:

(a) *Gauge theories* (classes **G**, **L**): finite shadow depth, collision residue with at most a simple pole. The bar complex is effectively quadratic (class **G**) or cubic (class **L**). The representation theory is governed by quantum groups with rational R-matrices.
(b) *Gravitational theories* (classes **C**, **M**): the shadow tower extends to degree 4 (class **C**) or is infinite (class **M**). The collision residue has higher-order poles. The bar complex is genuinely non-formal: the full A_∞-tower is irreducible. The Koszul duality $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ (or $\mathcal{W}_N^! = \mathcal{W}_N^{k'}$ at the dual level) exchanges matter and ghost sectors.

DS reduction is the functor connecting these regimes: it transports affine Kac–Moody (class **L**, gauge) to $\mathcal{W}$-algebras (class **M**, gravity) by raising the pole order through the Sugawara construction. The physical content: the Sugawara stress tensor converts gauge currents into a metric, and the bar complex detects this conversion as the appearance of higher collision poles.

3.4. **The critical string at $c = 26$.** The Virasoro Koszul duality $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ identifies three distinguished points:

c	**Property**	κ	**Clifford**
0	Gauge point: $\kappa = 0$, flat bar	0	degenerate
13	Self-dual point: $\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$	13/2	non-degenerate
26	Critical string: $\kappa^! = 0$	13	exterior algebra

At $c = 0$: $\kappa(\mathrm{Vir}_0) = 0$, the bar complex is flat at all genera, and the genus tower is trivial. At $c = 13$: $\kappa(\mathrm{Vir}_{13}) = 13/2 = \kappa(\mathrm{Vir}_{13}^!)$, the complementarity pairing is symmetric, and the entire shadow tower is self-dual. At $c = 26$: $\kappa(\mathrm{Vir}_0^!) = 0$, the Koszul dual is flat; the Clifford algebra of the complementarity pairing degenerates to an exterior algebra, and the matter-ghost cancellation condition $\kappa_{\mathrm{eff}} = \kappa(\text{matter}) + \kappa(\text{ghost}) = 0$ of the bosonic string is satisfied. The 26-dimensional critical string is the point where the Koszul dual's bar complex becomes flat: the gravitational content migrates from curvature to bar cohomology.

4. The Calabi–Yau bridge

Volumes I and II take a chiral algebra as given. The source of chiral algebras is geometry: a Calabi–Yau category C of dimension d produces a chiral algebra A_C on a curve X, with the CY trace realized as the modular characteristic. This is the content of Volume III.

4.1. The functor Φ. The construction proceeds in three steps. Given a CY_d category $\mathcal{C}$:

(1) The cyclic A_∞-structure on $\mathcal{C}$ determines a *Lie conformal algebra* $L_{\mathcal{C}}$: the cyclic bar differential gives the λ-bracket, the cyclic trace gives the invariant bilinear form, and the higher A_∞-operations give higher Lie conformal products.
(2) The *factorization envelope* $U_X^{\mathrm{fact}}(L_{\mathcal{C}})$ produces a factorization algebra on $\mathrm{Ran}(X)$, hence a chiral algebra $A_{\mathcal{C}}$ on X. The PBW theorem for factorization envelopes guarantees the expected filtration.
(3) The $\mathbb{S}^d$-framing on categorical Hochschild homology determines the operadic enhancement. For $d = 2$: the $\mathbb{S}^2$-action gives an E_2-structure with braided monoidal representation category. For $d = 3$: holomorphic Chern–Simons theory on $\mathcal{C}$ breaks E_2 to E_1, and the braided structure is recovered via the Drinfeld centre of the E_1-monoidal category.

Theorem 4.1 (CY-A_2). *The functor Φ is constructed for $d = 2$: all three steps are proved. The modular characteristic of $A_{\mathcal{C}}$ equals the Euler characteristic: $\kappa(A_{\mathcal{C}}) = \chi^{\mathrm{CY}}(\mathcal{C})$.*

For $d = 3$, steps (1) and (2) are unconditional; step (3) is a programme conditional on the chain-level $\mathbb{S}^3$-framing. The passage from $d = 2$ to $d = 3$ replaces a geometric braiding (from $\pi_1(\mathrm{Conf}_2(\mathbb{R}^2)) = \mathbb{Z}$) with an algebraic one (from the Drinfeld centre of the E_1-monoidal representation category): this is the CY analogue of the Tannakian reconstruction of a quantum group from its module category.

4.2. The E_1 stabilization. For $d \geq 3$, the CY chiral algebra is E_1, not E_d. On toric $GL(d)$-invariant local charts the relevant Schouten–Nijenhuis bracket on polyvector fields of $\mathbf{C}^d$ vanishes in the scalar channel; general CY categories are E_1 by the two-stage specialization and degree bookkeeping, not by Lie-level abelianity. The CoHA construction produces an ordered (associative) product from the extension correspondence $0 \to V' \to V \to V'' \to 0$, which has a preferred direction: sub before quotient. This is E_1, not E_2.

The framing obstruction lies in $\pi_d(BU)$, which is $\mathbb{Z}$ for even d and 0 for $d \equiv 1, 3, 7 \pmod 8$. The pattern repeats with period 8 by Bott periodicity. For $d = 2$: the obstruction is $c_1 \in \pi_2(BU) = \mathbb{Z}$, the braiding parameter. For $d = 3$: the obstruction vanishes ($\pi_3(BU) = 0$), so the $\mathbb{S}^3$-framing trivializes and the E_1-structure receives no additional enhancement. For $d = 4$: $\pi_4(BU) = \mathbb{Z}$ produces a shifted symplectic level (p_1), the CY_4 analogue of the Kac–Moody level.

4.3. The $K3 \times E$ prototype. The product of a K3 surface S and an elliptic curve E is the canonical CY_3 testing ground: simple enough to compute, rich enough to obstruct.

The lattice $\Lambda^{3,2}$ has signature $(3, 2)$; the Jacobi form $\phi_{0,1}(\tau, z)$ is the K3 elliptic genus; the Igusa cusp form Δ_5 is the Borcherds denominator identity. The Heisenberg-level modular characteristic of the chiral de Rham complex is $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3 \times E) = 3 = \dim_{\mathbf{C}}$, verified by six independent paths (additivity under Cartesian product: $2 + 1$; distinct from the total-space Hodge supertrace $\Xi(K3 \times E) = \chi(\mathcal{O}_{K3\times E})$, which is Künneth-multiplicative and vanishes for $K3 \times E$; see Volume III Remark **??** for the Route A / Route B adjudication). The BKM automorphic weight is a distinct quantity: $\kappa_{\mathrm{BKM}} = 5$ (the weight of Δ_5).

These are different numbers measuring different things. A single CY_3 admits multiple chiral algebraizations, each producing a different modular characteristic. The $\kappa_\bullet$-spectrum

$$\mathrm{Spec}_{\kappa_\bullet}(K3 \times E) \;=\; \{0, 3, 5, 24\}$$

records four total-space and fibre-normalized values: $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3\times E}) = 0$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$ (the complex dimension, Route B), $\kappa_{\mathrm{BKM}} = 5$ (the Borcherds weight), and $\kappa_{\mathrm{fiber}} = 24$ (the K3 lattice rank). The value $\chi(\mathcal{O}_{K3}) = 2$ belongs to the K3 fibre lane; it is not $\kappa_{\mathrm{cat}}(K3 \times E)$.

Four structural obstructions separate the shadow obstruction tower from the full automorphic form:

(O1) *Categorical*: the shadow tower produces Taylor coefficients; the denominator identity is an automorphic form on $\mathcal{H}_2$.

(O2) $\kappa_\bullet$*-mismatch*: $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}$.

(O3) *Second quantization*: the physical DT partition function is the OP-reduced $K3 \times E$ invariant, with the elliptic translation factor removed, and is the DMVV symmetric product of single-copy data; the bar complex is the single copy.

(O4) *Schottky*: at genus $g \geq 4$, the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ is no longer dominant, and the shadow tower is imprisoned in tautological classes.

Each obstruction points to a research programme. The $\kappa_\bullet$-spectrum is the first invariant that separates CY chiral algebras with the same underlying manifold.

4.4. **E_1 at $d = 3$ via holomorphic Chern–Simons.** For a CY threefold X, holomorphic Chern–Simons theory on X with gauge group G produces a chiral algebra A_X of E_1-type. The spectral parameter of the R-matrix is identified with the holomorphic coordinate of the 4d theory restricted to a 2d boundary. The braided monoidal structure on the representation category is recovered through the Drinfeld centre:

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(A_X)\big) \;\simeq\; \mathrm{Rep}^{E_2}(A_X)\,,$$

where the right-hand side is the braided monoidal category of E_2-modules. For toric CY_3, the E_1-algebra is the critical CoHA $\mathrm{CoHA}(Q_X)$, identified with the positive half $Y^+(\widehat{\mathfrak{g}}_{Q_X})$ of an affine super Yangian. For $\mathbf{C}^3$, the Jordan quiver gives $\mathrm{CoHA}(\mathbf{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$; the full $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$ appears only after the Drinfeld double or the corresponding evaluation construction.

The dimensional reduction principle underlies the full trilogy: $4d \to 3d$ forgets the braiding ($E_2 \to E_1$); $3d \to 2d$ forgets the ordering ($E_1 \to E_\infty$ by Σ_n-coinvariance). The five main theorems live at the $2d$ level; the seven faces of the collision residue live at the $3d$ level; the CY source lives at the $4d$ level. The modular Koszul engine converts enumerative geometry (Donaldson–Thomas invariants, root multiplicities, denominator identities) into homotopy algebra (bar complexes, MC elements, genus expansions). One Maurer–Cartan element Θ_{A_C} absorbs the entire automorphic datum.

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